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Site-Specific Assessment of CO₂ Storage Potential in Chalk: The Harald East Experience

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Abstract

This research provides a site-specific assessment of the viability and safety of long-term CO₂ storage within chalk formations, utilizing the Harald East Field in the Danish Central Graben as a case study. By leveraging over three decades of production data, 4D seismic monitoring, and subsidence records, we established a multiphysics modeling framework to evaluate injectivity, storage capacity, containment, and geomechanical stability. The simulations suggest that the highest capacity for CO₂ storage, limited by the pre-depletion pressure of the reservoir, can be attained while ensuring that the structural and stratigraphic integrity of the plume is preserved during both the injection and post-injection stages. Geomechanical modeling indicates a minimal risk of fault reactivation, even under stress regime transitions induced by repressurization. While chalk's low permeability poses injectivity challenges, its high porosity enables substantial storage potential, provided injection strategies are carefully optimized. Although the low permeability of chalk presents challenges for injectivity, its high porosity allows for significant storage potential, contingent upon the careful optimization of injection strategies. Reactive transport modeling indicates localized porosity reduction near the injection wells due to rock-fluid chemical interactions; however, injectivity is not compromised due to the extended horizontal well sections. In the caprock, long-term geochemical reactions, especially under scenarios involving impure CO₂ injection, facilitate secondary mineral precipitation, which enhances sealing capacity. Diffusion simulations conducted over a 10,000-year timeframe confirm minimal migration of CO₂ and impurities into the caprock, with porosity reductions attributed to mineral trapping and selfsealing processes. While thermal effects are spatially constrained, they do affect fluid behavior and reaction kinetics in proximity to the wellbore, and the cooling effects on the caprock merit further investigation. In summary, the findings affirm the technical feasibility of CO₂ storage in depleted chalk reservoirs and provide a transferable modeling framework for assessing analogous carbonate systems, thereby contributing to a more comprehensive understanding of CO₂ dynamics in formations characterized by low permeability and high porosity.

Introduction

Chalk formations, especially those located in the North Sea, are increasingly regarded as potential sites for subsurface CO₂ storage due to their significant pore volume, well-established production history, and pre-existing infrastructure (Bonto et al. 2021). Nevertheless, chalk presents a distinct array of challenges that set it apart from more frequently examined storage lithologies like sandstones or basalts.

Chalk is a soft, fine-grained carbonate rock known for its high porosity (often surpassing 40%) and very low permeability (generally below 10 mD). This unique combination renders it both advantageous and problematic for CO₂ storage. On one hand, the high porosity (if maintained post-depletion) provides ample pore space for CO₂ injection. Conversely, the low permeability introduces considerable injectivity challenges, often requiring extensive horizontal wells or stimulation techniques to achieve acceptable injection rates.

From a mechanical perspective, chalk is highly deformable (Boade et al. 1989). During hydrocarbon extraction, it experiences poroelastoplastic compaction, with vertical displacements exceeding several meters in depleted fields. This deformation modifies the in-situ stress regime and can influence fault stability, well integrity, and the sealing capacity of caprock. These mechanical alterations must be accurately considered when assessing the viability of CO₂ injection into depleted chalk reservoirs.

Geochemically, chalk exhibits reactivity. It comprises minerals such as calcite, feldspars, and smectites that may dissolve or precipitate in the presence of CO₂-rich brine (Mamonov et al. 2025). Such reactions can result in alterations to porosity and permeability, potentially enhancing or undermining storage integrity (Amour et al. 2025b). Additionally, the presence of impurities in the injected CO₂ stream (e.g., H₂S, SO₂, NO₂) complicates the chemical environment, particularly in the caprock, where longterm diffusion and mineralization processes dictate containment (Xu et al. 2007). The combined dissolution and precipitation reactions in the reservoir or caprock upon CO₂ exposure determine the mineralogical variations and resulting alterations in rock properties such as porosity (Tambach et al. 2015). These risks associated with geochemical interactions are evaluated based on the extent of mineralogical variations and transport of CO₂ and accompanying impurities.

Despite these complexities, chalk formations are still inadequately represented in the literature concerning CO₂ storage (Bonto et al. 2021). Most of the current research emphasizes siliciclastic reservoirs or deep saline aquifers, thereby creating a significant knowledge void regarding the behavior of CO₂ in carbonate systems, especially those that have experienced considerable depletion.

The Harald East Field, situated in the Danish Central Graben, presents a unique opportunity to fill this void. It is a depleted gas reservoir with a wealth of historical data, encompassing over 30 years of production history, 4D seismic monitoring, platform subsidence measurements, and core and log data from various wells. This paper offers a site-specific evaluation of CO₂ storage potential in a depleted chalk reservoir, employing Thermo-Hydro-Mechanical-Chemical (THMC) analyses. The study examines the injectivity, capacity, and containment of CO₂; evaluates the geomechanical stability of faults and the risk of induced seismicity; investigates geochemical interactions between CO₂ (including impurities) and the reservoir-caprock system; and assesses the longterm integrity of the caprock under diffusive transport conditions. The results yield a validated modeling framework for evaluating analogous chalk reservoirs in the North Sea and other carbonate-rich basins.

Methodology

This work utilizes a multiphysics analysis framework to assess the viability and safety of CO₂ storage within the Harald East chalk reservoir. The methodology incorporates both field-scale and fault-scale simulations, reactive transport modeling, and history matching, leveraging an extensive dataset accumulated over more than thirty years of hydrocarbon extraction. The modeling approach is crafted to encapsulate the intricate

interactions among fluid dynamics, mechanical deformation, thermal variations, and geochemical processes that dictate the behavior of CO₂ in chalk formations.

The reservoir model is founded on a high-resolution geological framework developed in Petrel, encompassing a lateral domain of approximately 9.5 km by 7.0 km and a vertical extent of 3.7 km (Figure 1). It comprises the chalk reservoir, overburden, underburden, and lateral boundaries. The primary storage layer is the D1 unit in the Ekofisk Fm, a reworked chalk formation composed of a mixture of Maastrichtian and Danian sediments and noted for its high porosity and low permeability. The caprock is formed by marl and shale layers that exhibit significantly lower permeability, thereby providing an effective seal. Material characteristics for each stratigraphic unit are determined based on core samples, well logs, and laboratory tests, including triaxial tests and PVT analyses.

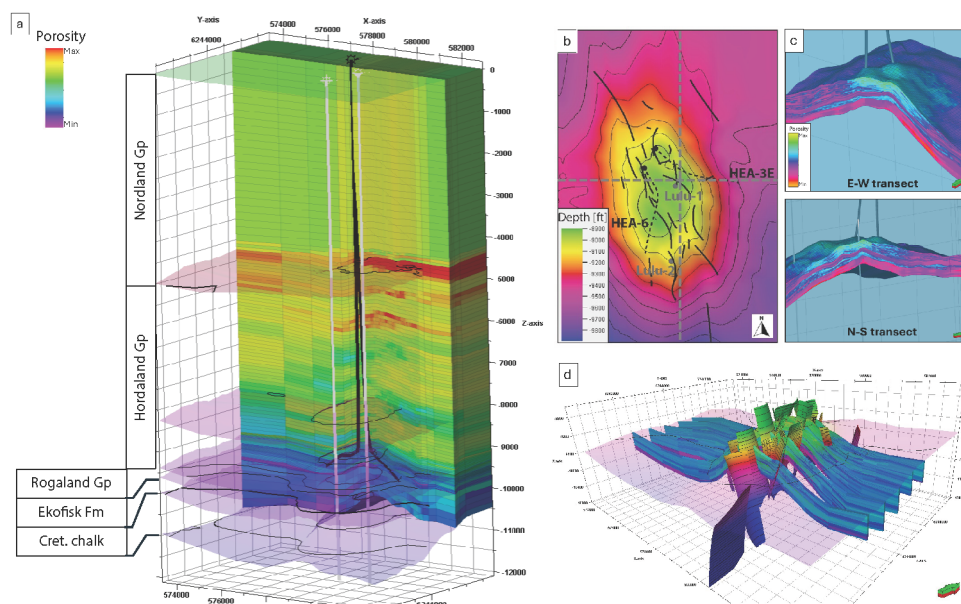


Figure 1—a) 3-D view of the geological model including the underburden (below Ekofisk Fm), the storage unit referred to as the Ekofisk Fm, and the 2.7 km-thick overburden up to the seabed. b) Map view of the faulted (black solid line) anticline structure in the Harald East area with the two production wells (HEA-3E and HEA-6) and the two exploration wells (Lulu-1 and Lulu-2) now plugged and abandoned. c) E-W and N-S transects of the Ekofisk Fm along the anticline structure showing the porosity distribution. The locations of the transects are shown in (b). d) 3-D view of the distribution of faults intersecting the reservoir-caprock interface, together with multiple E-W transects displaying porosity. Note that the faults located in the underburden and overburden are not displayed here because the simulation results have suggested a negligible risk of reactivation.

Dynamic reservoir simulations are conducted using SLB Intersect, which resolves non-isothermal multiphase flow equations. These simulations are integrated with the Visage geomechanics simulator to incorporate stress evolution, deformation, and fault dynamics (Figure 2). A two-way coupling scheme is implemented, allowing for the iterative exchange of pressure, temperature, and saturation fields between the flow and geomechanical solvers until convergence is reached at each time step. The Thermo-HydroMechanical (THM) model is calibrated against historical production data, 4D seismic observations, and platform subsidence measurements, ensuring that the simulated mechanical response accurately mirrors the depletion history of the field (Hosseinzadeh et al. 2025).

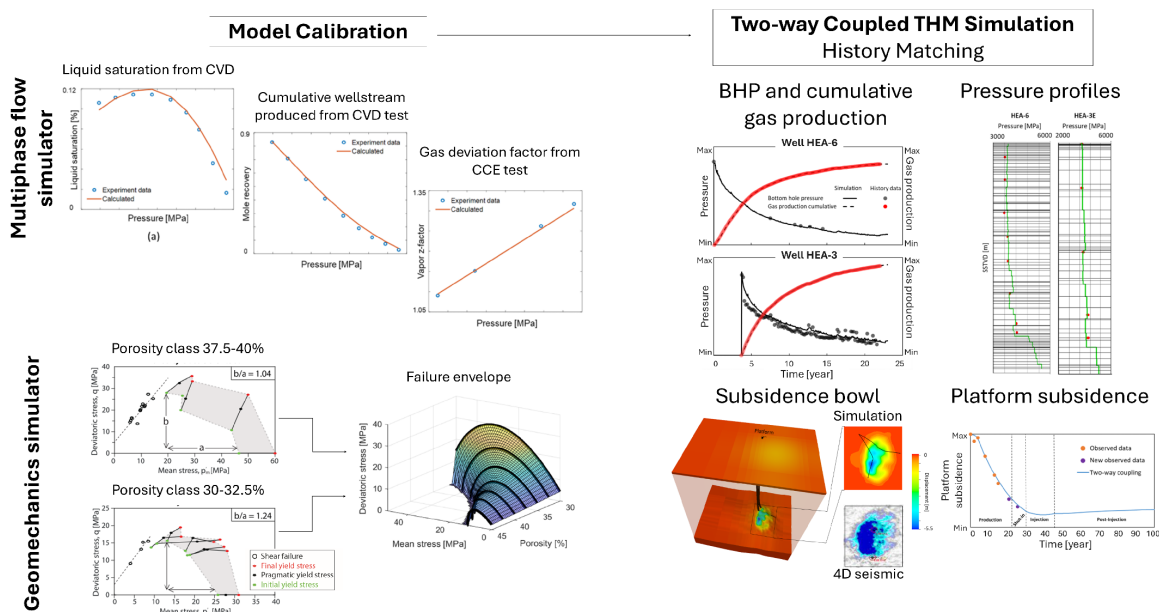


Figure 2—Workflow of the THM simulation. The model calibration encompasses i) PVT characterization of the hydrocarbon phases using constant volume depletion (CVD) tests and constant composition expansion (CCE) tests to tune the EOS model and ii) assessment of the elastic, tensile, shear, plastic, and creep properties of reservoir chalks to reconstruct their failure envelopes. Then, the THM simulation results are history matched against bottomhole pressure (BHP), formation pressure test, production data, as well as 4D seismic and platform subsidence data.

To evaluate the potential for fault reactivation and induced seismicity during repressurization of the depleted reservoir through CO₂ injection, a fault network model has been constructed in Petrel. The model comprises over 100 faults, grouped into three categories: (i) faults located exclusively in the overburden, (ii) faults intersecting the caprock-reservoir boundary, and (iii) faults located below this boundary. This field-scale model has identified one critically stressed fault through slip tendency analyses. Subsequently, a frictional model focused on this critically stressed fault is built in Abaqus, incorporating the surrounding reservoir and caprock units. The fault is represented using a frictional contact interface, with parameters obtained from laboratory experiments and existing literature, while the applied pore pressure, temperature, saturation, and stress evolutions are extracted from the field-scale model. Various scenarios in terms of friction coefficients (0.27 to 0.6) and discretization methods (coarse with brick elements and fine with tetrahedral elements) are considered to assess slip behavior and estimate potential seismic event magnitude.

Reactive transport modeling is performed using PHREEQC, concentrating on the interactions among CO₂ (including impurities such as H₂S, SO₂, and NO₂), brine, and rock minerals. The simulations take into account both equilibrium and kinetic reactions, considering mineral dissolution, precipitation, and alterations in porosity. The geochemical model is applied to both the reservoir and caprock, with a specific focus on the long-term diffusion of CO₂ and its impurities into the seal. Equilibrium interactions between the rock, brine, and gas phases are considered for the reservoir, while a kinetic approach is applied to account for the geochemical reactions within the caprock. The mineralogy of caprock includes reactive components such as feldspars, smectites, and carbonates, which affect the degree and nature of mineral trapping and self-sealing mechanisms. The transport of CO₂ and impurities has been investigated through a multicomponent diffusive process considering both the electromigration and diffusion of the aqueous species. The diffusion extent of CO₂ and impurities into the caprock is explored based on the integrated rock-fluid chemical interactions, which affect the diffusion coefficient dynamically. The transport calculations in the caprock are performed for 10,000 years, as the geological timescale.

To assess the impact of chemical reactions on fluid injectivity, a reactive transport simulation is conducted utilizing a twodimensional cross-sectional slice ((Figure 3) derived from the detailed three-dimensional

geological model of the Harald East site (Figure 2), which measures 2.5 km in length, 60 meters in width, and 200 meters in depth. The simulation utilizes the SLB Intersect platform, a history-matched thermal flow model, to accurately represent the dynamics of non-isothermal multiphase flow and the movement of chemical constituents, including hydrocarbon solubility and dryout phenomena. Reactive processes such as mineral precipitation and dissolution are calculated every 10 days using PHREEQC, facilitating a time-resolved integration between fluid flow and geochemical changes within the subsurface.

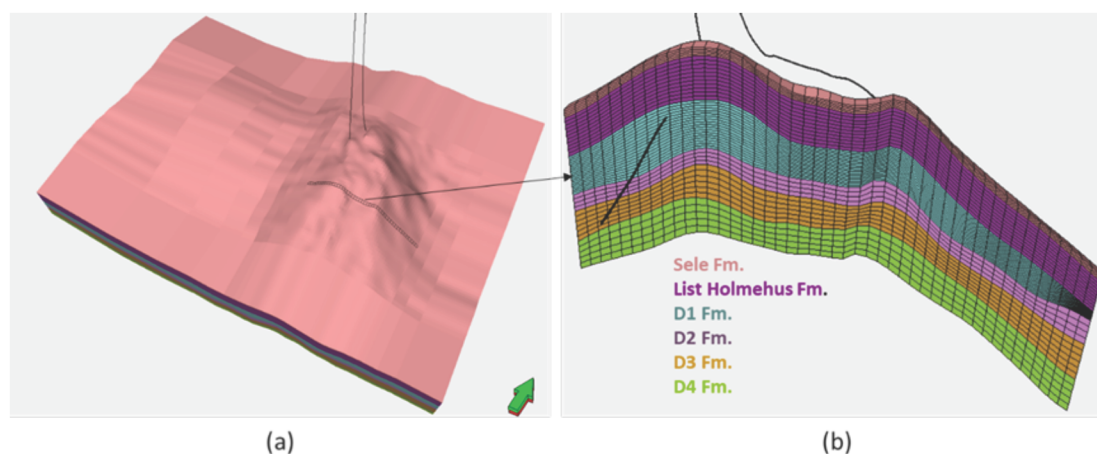


Figure 3—a) 3D representation of the reservoir and caprock, b) 2D cross-sectional model extracted from the 3D reservoir, including both the reservoir and part of the caprock.

The injection scenario considers that cold CO₂ will be injected over a 15-year timeframe, aiming to restore reservoir pressure to pre-production levels. Through iterative simulations, a safe range of injection rates and bottom-hole pressures was established to ensure that reservoir pressure remains below the virgin reservoir pressure, the fracture initiation pressure, and thresholds that could induce fault slip or compromise caprock integrity. The simulations assess a variety of operational and geological uncertainties, including fluctuations in fault friction, impurity levels, and thermal gradients, to evaluate the resilience of the storage system.

This modeling technique facilitates the examination of CO₂ injection and storage dynamics within chalk formations by considering the interrelated thermal, hydraulic, mechanical, and chemical processes that affect injectivity, containment, and the progression of the system over time. Although initially designed for the Harald East Field, this methodology is versatile and can be modified for application in other carbonate reservoirs that share comparable geological and operational traits.

Results

Storage capacity

During the gas extraction phase, the most substantial compaction occurs within the highly porous zones of the chalk formation. Areas exhibiting porosity greater than 38% experience vertical viscoplastic strains exceeding 10% on average. In contrast, as porosity decreases toward the flanks of the anticline structure, the extent of compaction diminishes significantly. Rocks with porosity below 30% undergo minimal to no permanent deformation. This compaction is primarily driven by reservoir pressure decline, which elevates the effective stress and preferentially compresses the weakest, most porous chalk layers. Consequently, the pore space - and thus the reservoir's capacity to store fluids - is reduced. Simulation results indicate that this compaction leads to an approximate 10% loss in pore volume, a factor that must be carefully considered when evaluating the feasibility of CO₂ storage in depleted, deformable chalk reservoirs.

The simulation results presented in Figure 4 demonstrate that thermal effects have a negligible influence on CO₂ storage capacity. This is attributed to the localized nature of thermal changes, which are confined

to the vicinity of the injection well. While cold CO₂ injection may affect caprock integrity and fault reactivation, these impacts are contingent upon the extent of cold temperature propagation and do not significantly alter the overall storage capacity. In contrast, mechanical effects such as compaction play a critical role. As illustrated in the figure, neglecting mechanical impacts leads to a substantial overestimation of CO₂ storage capacity due to the slower pressure build-up. The pressure changes over time shows a marked difference between scenarios with and without compaction, regardless of injection temperature.

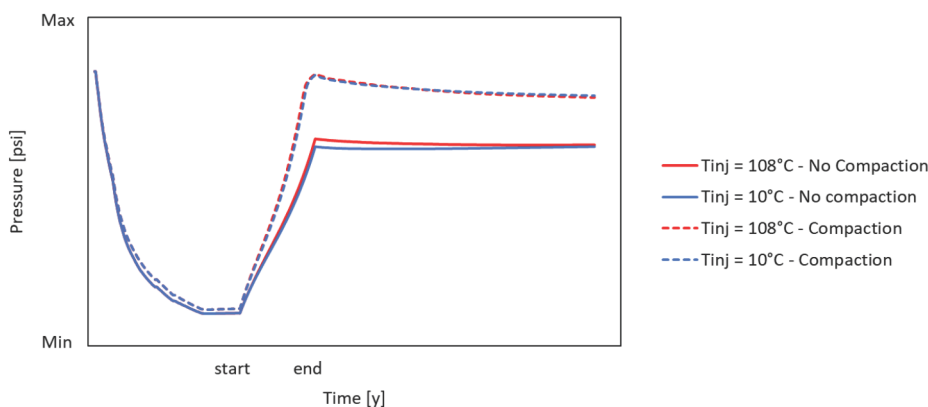


Figure 4—Pressure response over time for different CO₂ injection temperatures with and without compaction.

Containment

The THM simulation extends decades beyond the end of injection to evaluate potential leakage through the caprock or toward the flanks of the anticline. The results predict the formation of two distinct CO₂ plumes during the initial years of injection, each associated with one of the two former production wells converted to injectors. As the injection progresses, these plumes coalesce into a single larger plume. During the post-injection stage, only minimal changes in CO₂ saturation are predicted in the crest region (Figure 5). Furthermore, lateral propagation toward the flanks of the anticline is negligible, indicating rapid stabilization of the injected CO₂. The CO₂ remains securely contained within the designated storage area, above the spill point located on the western side of the anticline.

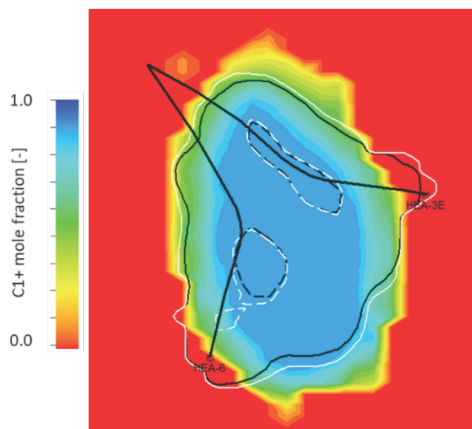


Figure 5—The movement of CO₂ during cold injection is illustrated through contour mapping. Black contours delineate the final extent of the CO₂ plume at the conclusion of the injection phase, while white contours mark its spread at the end of the post-injection period. Dashed lines indicate plume behavior near the crest of the structure, whereas solid lines correspond to the flanks. The underlying color gradient on the map represents the initial distribution of heavier hydrocarbon components (C1+) at the onset of production, offering insight into the spatial dynamics of the injected CO₂ relative to existing reservoir fluids.

Containment is further ensured by a thick caprock overlying the reservoir. In this area, the Rogaland Group is approximately 60 m thick (Figure 1) and consists of shale-dominated sediments from four distinct formations, from base to top: the North Sea marls (also called Vale Fm), and the Lista, Sele, and Balder Fms. Besides, mercury injection capillary pressure data from the latter formations collected in this study and published in literature (Worden et al. 2020) indicate that the CO₂ height column that the seal interval can hold ranges between 140 m and 470 m, with a mean value around 315-335 m. Within the Harald East area, the maximum CO₂ column height between the crest and the spill point is equal to 141 m. The risk of membrane sealing failure is thus deemed low in the study area. Finally, approximately 1.2 km of shale-dominated sediments from the Hordaland Group (Figure 1) with a porosity between 20% and 45% overlies the Rogaland Group, providing additional waste zones that can mitigate potential leakage by storing CO₂ in their pore space.

Field deformation and fault stability

The geomechanical analysis, executed through a two-way coupled methodology at the field scale between flow and mechanical solvers, documented the changes in stress and deformation throughout both the production and injection phases. The findings indicate that the reservoir underwent considerable compaction during depletion, with vertical displacements surpassing 5 meters in certain regions dominated by porous and mechanically weak chalk (Figure 2). During the injection phase, the reservoir progressively repressurizes, and the stress path is locally influenced by the chalk's deformation regime (Figure 6). In regions dominated by elastic deformation during depletion, the stress path is nearly non-hysteretic. Repressurization restores the stress state, moving faults away from critical conditions. However, the plastic compaction of chalk maintains high effective horizontal stress during depletion due to the lateral expansion of the rock. As formation pressure increases toward the end of the injection phase, the THM model predicts that the vertical stress becomes smaller than the horizontal stress, thereby shifting the faulting regime from normal to reverse conditions. This transition is particularly significant for assessing fault stability, as it modifies the slip tendency on pre-existing faults. Recent numerical simulation studies have suggested that subsurface rock compaction can lead to a change in faulting regime during repressurization and subsequent deterioration of fault stability (Jeanne et al. 2020; Seth et al. 2024; Xin et al. 2024).

To assess the likelihood of fault reactivation, a fault-scale model was constructed around the critically stressed fault identified in the field-scale model (Figure 6). This model included a frictional contact interface and chalk-specific constitutive behavior, which encompassed rate-dependent plasticity and pore collapse. Simulations conducted with varying friction coefficients revealed that for realistic values (> 0.36), the slip tendency remained below critical limits throughout the injection period. Only under conservative assumptions (e.g., a friction coefficient of 0.27) did localized slip occur, and even in such cases, the estimated moment magnitude stayed below 2.2, suggesting a low risk of induced seismicity, which is a parameter that can be operationally controlled should it be required.

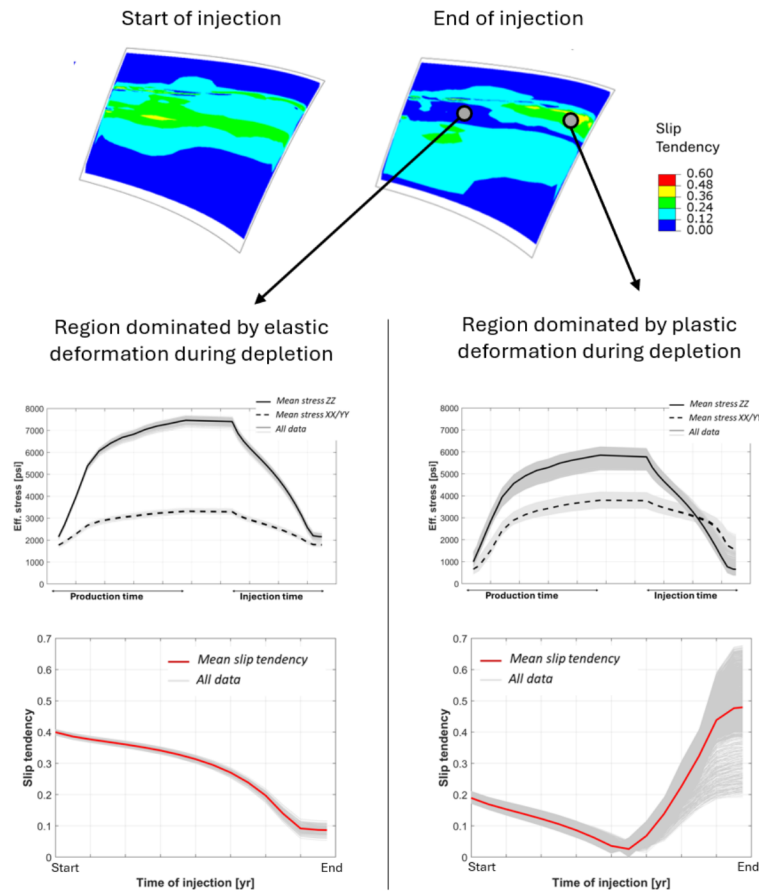


Figure 6—3-D view of a critically stressed fault plane showing the evolution of slip tendency from the start to the end of CO₂ injection (Top). In regions of the fault surrounded by chalk, which predominantly deforms elastically during depletion, the stress state is nearly restored to virgin conditions (middle), resulting in mechanical stabilization of the fault plane during injection (decrease in slip tendency value, bottom). However, the plastic compaction of chalk during depletion caused stress hysteresis that might lead to fault reactivation toward the end of the injection phase.

Geochemical Alterations and Caprock Integrity

To assess the mineralogy and porosity alterations in the reservoir, the precipitation and dissolution reactions are modeled using an equilibrium approach. We considered thermodynamic simulations for three cases: 1- The reservoir is in the initial condition containing wet gas at high pressure and temperature; 2- The reservoir is depleted during hydrocarbon production, causing the pressure to be decreased; 3- CO₂ is injected to the reservoir after depletion and the pressure is increased back to the level at the initial conditions (before the production stage). Due to uncertainties in the reservoir mineralogy, we considered four scenarios (a to d) and performed simulations for each. As the initial equilibrium of the reservoir is disrupted by depletion and subsequent CO₂ injection, changes in mineralogy were examined during both stages, as illustrated in Figure 7. The associated porosity changes resulting from mineral dissolution and precipitation are also evaluated and presented in the same figure. The results for pure CO₂ injection indicate that porosity changes in the host reservoir rock are minimal. Even in the presence of impurities, porosity remains largely unaffected, as dissolution processes are largely offset by concurrent precipitation reactions.

After depletion			After CO ₂ injection			Scenario
Dissolves -	Precipitates +	Porosity change	Dissolves -	Precipitates +	Porosity change	
Quartz, Kaolinite	Calcite	↓ 0.0006 P.U	Calcite	Quartz, Kaolinite	↑ 0.009 P.U	a
Quartz, Kaolinite, Calcite	Smectite	↑ 0.00013 P.U	Smectite, Illite	Quartz, Kaolinite, Calcite, Dawsonite, Siderite, Pyrite	↓ 0.213 P.U	b
Quartz, Kaolinite	Calcite	↓ 0.0003 P.U	Calcite	Quartz, Kaolinite	↑ 0.0045 P.U	c
Smectite	Quartz, Kaolinite, Calcite, Illite	↓ 0.0005 P.U	Smectite, Illite	Quartz, Kaolinite, Calcite, Dawsonite, Siderite	↓ 0.198 P.U	d

Figure 7—Geochemical modeling results showing changes in mineralogy and porosity in the reservoir units of the Harald East field after depletion and subsequent pure CO₂ injection. Porosity changes are reported in porosity units (P.U).

We considered two scenarios for the brine diffusing into the seal. In the first scenario, the diffusive brine is equilibrated with the reservoir rock before diffusion. This reduces the acidity of the diffusive brine due to reactions with reservoir minerals. In the second scenario, as the worst-case condition, the diffusion occurs without prior equilibration, leading to a more acidic brine to diffuse. In both cases, the long-term diffusion simulations conducted over a 10,000-year period revealed that CO₂ and its impurities (H₂S, SO₂, and NO₂ each in 0.1 vol.%) migrate a few meters into the seal. In the first scenario (Figure 8a), a continuous porosity reduction and sealing are seen due to the dominance of precipitation reactions and CO₂ chemical trapping. In the second scenario (Figure 8b), a narrow dissolution zone develops at the reservoir-caprock interface, followed by a wider precipitation zone where minerals such as silicates and carbonates are formed, which effectively diminishes porosity and enhances the sealing capacity. The presence of Fe-, Ca-, and Mg-rich feldspars within the caprock plays a significant role in this self-sealing mechanism, even under adverse impurity conditions.

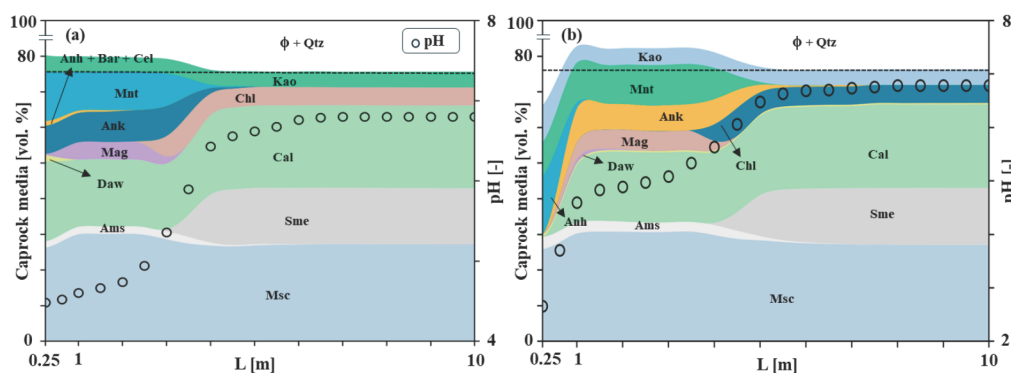


Figure 8—10,000-year alteration in Harald East caprock mineralogy following the diffusion of CO₂, considering a) prior equilibration and b) no prior equilibration of the diffusive brine with the reservoir minerals. L represents the length above the reservoir-caprock interface. The abbreviations in the figures are: Mnt: Montmorillonite, Msc: Muscovite, Sme: Smectite, Cal: Calcite, Ank: Ankerite, Daw: Dawsonite, Kao: Kaolinite, Qtz: Quartz, Mag: Magnesite, Anh: Anhydrite, Ams: Amorphous silica, Chl: Chlorite. is the porosity of caprock.

Thermal effect

The thermal effects associated with the injection of cold CO₂ were also captured in the simulations (Figure 9). Near the injection well, there was a notable drop in temperatures, which affected fluid properties and reaction kinetics. Nevertheless, the thermal front remained localized, and its influence on the overall reservoir and caprock system was minimal.

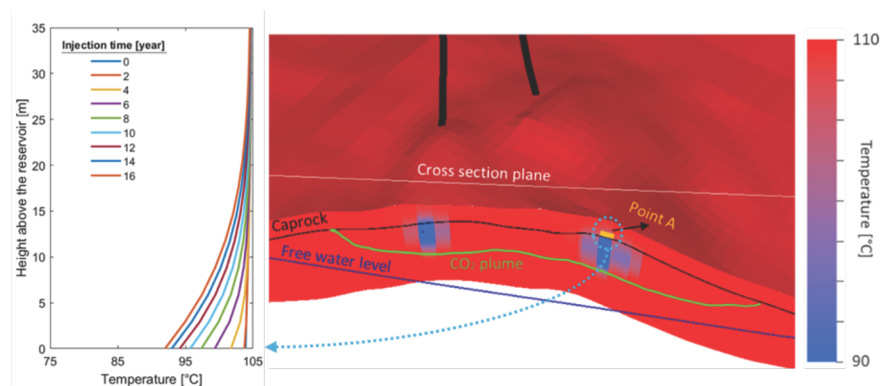


Figure 9—E-W transect showing the temperature propagation around the two injectors (left). The reader is referred to Figure 1 for the location of the transect. Prediction of the cooling by diffusion in the caprock at point A located above the injector well HEA-3E (right).

Injectivity

Injecting pure CO₂ into a depleted gas reservoir with ~30% water saturation near the well initiates CO₂ dissolution in water, forming carbonic acid and slightly lowering pH. This may cause minor mineral dissolution, with porosity changes under 0.1%, mostly near the well (Figure 10a). Dissolution is more significant on the flanks, where water saturation is higher. As CO₂ replaces residual gas, reservoir pressure may increase, enhancing structural integrity. Low water content at the crest limits reactions and reduces hydrate formation risk (Fernandes et al. 2024). Figure 10b shows the CO₂ mole fraction in water peaks at 1.5% near the well, while the temperature drops from 106°C to 30°C. Water vapor in the gas phase decreases from 2% to 0.5% over time due to cooling and pressure increase.

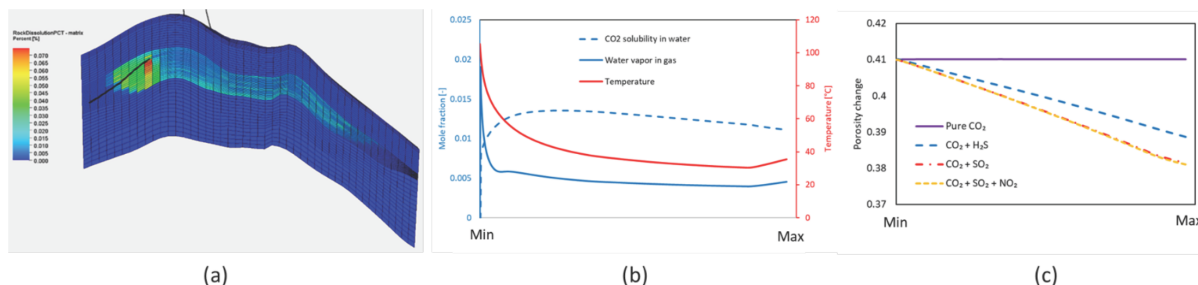


Figure 10—(a) Changes in porosity resulting from the injection of pure CO₂ are visualized, highlighting the spatial impact of geochemical interactions. (b) For the grid cell adjacent to the injection well, the evolution of key parameters is shown: the mole fraction of CO₂ dissolved in the aqueous phase (dashed blue line), the concentration of water vapor in the gas phase (solid blue line), and the temperature profile (solid red line), all under pure CO₂ injection conditions. (c) Temporal variations in porosity are compared across multiple injection scenarios for a cell hydraulically connected to the well, illustrating the sensitivity of rock structure to different operational strategies.

Unlike pure CO₂ injection, which primarily causes rock dissolution, injecting impure CO₂ can also lead to mineral precipitation. When CO₂ mixed with impurities such as H₂S, SO₂, and NO₂ is injected into a depleted gas reservoir with approximately 30% water saturation, the resulting chemical processes become significantly more complex (Xu et al. 2004). These impurities dissolve in the residual water and form stronger acids, such as sulfurous acid (H₂SO₃) and nitric acid (HNO₃), in addition to carbonic acid. The combined acidity lowers the pH more drastically, enhancing mineral dissolution and accelerating changes in the rock matrix.

Moreover, H₂S can react with metal ions to form metal sulfides, which may clog pore spaces, while SO₂ and NO₂ increase the likelihood of solid phase formation, further reducing permeability (Pearce et al. 2019). In scenarios involving these impurities, the porosity near the well decreases more significantly compared to pure CO₂ injection. In particular, the presence of all three impurities (H₂S, SO₂, and NO₂)

can reduce porosity by up to 7%, dropping from an initial 41% to 38% (Figure 10c). A scenario with only H₂S leads to a smaller porosity reduction, indicating that combined acidic effects have a greater impact on reservoir structure.

One of the main advantages of using depleted hydrocarbon reservoirs is the ability to utilize existing infrastructure, such as predrilled wells with high reservoir contact (originally designed for maximum productivity). In the Harald East field, two long horizontal wells (each with ~1500 m of perforated intervals) are already available and considered for CO₂ injection. For the planned injection rate, simulations show that the combined injectivity of these two wells is significantly higher than required. To confirm that this capacity remains adequate under adverse conditions, several hypothetical failure scenarios were tested: complete loss of one well, gradual injectivity decline (e.g., from mineral precipitation), loss of high-permeability intervals due to mechanical collapse, and overall reduced injectivity at the start of injection (after years of production). In all cases, the available injectivity still exceeds the designated injection rate, indicating a comfortable operational margin for the project as currently designed. It should be noted that this conclusion applies to the planned rate; if substantially higher injection rates were required in the future, the system's resilience to well failures would need to be re-evaluated. Figure 11 shows the CO₂ injection rate and cumulative injected volumes for the base case and each failure scenario.

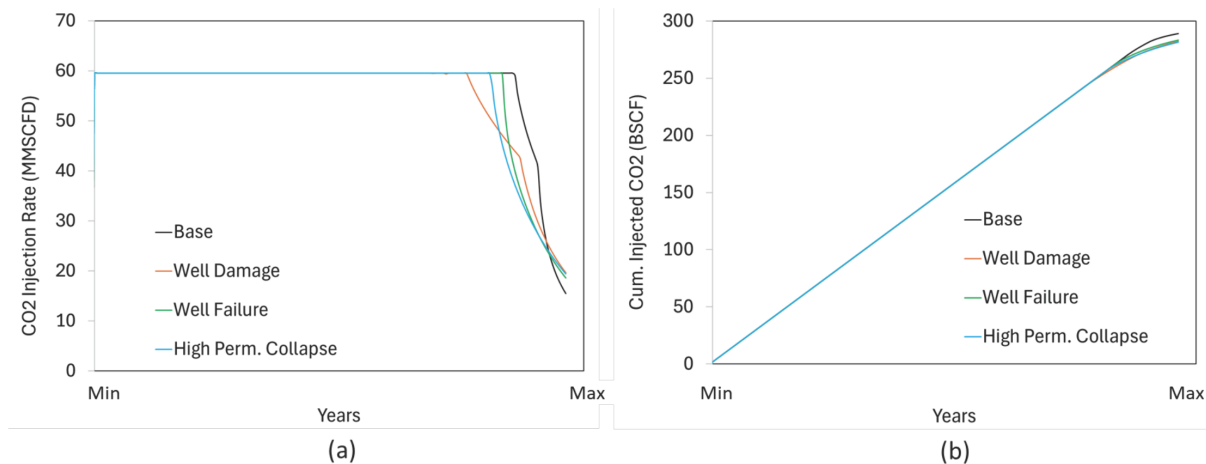


Figure 11—a) CO₂ injection rate, and b) Cumulative injected CO₂ for some of the injection failure scenarios

In addition to the above-mentioned scenarios, we conducted several simulations to predict the operational bottom-hole and average reservoir P-T conditions. As Figure 12 shows, these conditions remain well outside the hydrate stability envelopes. Therefore, hydrate formation is not anticipated under the expected reservoir conditions, whether considering pure CO₂, the in-situ reservoir hydrocarbons, or their mixtures.

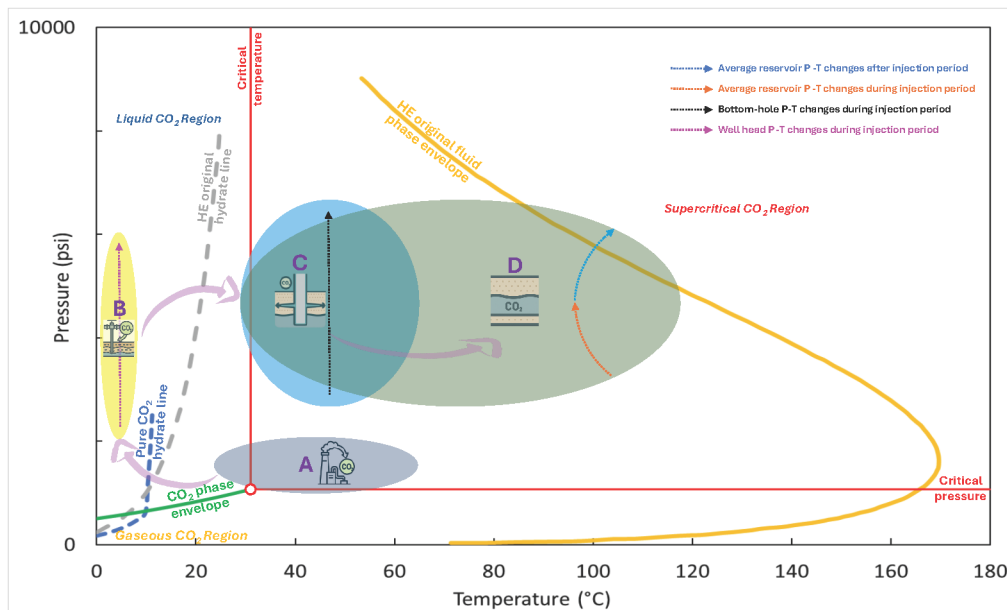


Figure 12—Pressure-temperature (P-T) phase envelopes for the Harald East reservoir fluid and CO₂, including the corresponding hydrate stability boundaries. For reference, the pure CO₂-water hydrate equilibrium line is shown. The circle around the CO₂ critical point highlights the uncertainty associated with impurities in the captured stream that may slightly shift the phase boundaries. During CO₂ injection, the real fluid behavior falls between that of the Harald East original reservoir fluid and pure CO₂. The shaded ovals labeled A, B, C, and D represent ranges of conditions: A corresponds to CO₂ at capture and transport conditions, B to wellhead conditions, C to bottomhole conditions, and D to reservoir conditions. At the wellhead (B), hydrate formation is not expected because the CO₂ stream is very dry. Similarly, the reservoir conditions (C and D) lie outside the hydrate stability region, indicating that hydrate formation is not anticipated under these conditions. It should be noted that the phase diagrams do not fully capture the behavior in region D, where the reservoir contains a mixed fluid. The intention is to depict only pressure and temperature conditions, not the full complexity of the mixture.

Discussion

The findings derived from the Harald East Field simulations offer an in-depth perspective on the interconnected physical and chemical mechanisms that regulate CO₂ storage within chalk formations. These results not only corroborate but also expand upon earlier research concerning CO₂ storage in carbonate systems (Madland et al. 2006; Liteanu et al. 2013), while simultaneously emphasizing the distinct challenges and benefits linked to depleted chalk reservoirs.

A pivotal insight from this investigation is the necessity of considering production-induced deformation when assessing fault stability during CO₂ injection (Amour et al. 2025a). The transition from a depleted state to a re-pressurized condition significantly modifies the stress regime, a phenomenon also observed in other geomechanical analyses of faulted reservoirs. In the context of Harald East, the transition from normal to reverse faulting conditions during injection was effectively captured through coupled THM simulations, revealing that the risk of fault reactivation remains low under realistic frictional parameters. This observation aligns with the findings of Khan et al. (2020), who illustrated that fault reactivation is influenced by factors such as reservoir deformation and size, boundary conditions, and injection strategies.

The inherent low permeability of chalk is a well-recognized obstacle to injectivity. This research substantiates that, despite the high porosity, the millidarcy-scale permeability of the Harald East reservoir demands meticulous well placement and pressure management. Comparable conclusions have been reached in other studies focused on carbonate storage, where challenges related to injectivity were mitigated through horizontal drilling or stimulation techniques (Ajayi et al. 2019). Nevertheless, the high porosity of chalk, if maintained after depletion, continues to represent a considerable advantage regarding storage capacity.

Geochemical modeling conducted in this research indicated that rock-fluid chemical interactions are spatially constrained within the reservoir, yet they are more significant in the caprock, especially under

scenarios involving impure CO₂ injection. The emergence of secondary minerals such as gypsum and anhydrite in proximity to the injection well, along with silicates and carbonates in the caprock, aligns with findings from other reactive transport investigations in carbonate and shale systems. These reactions may locally diminish porosity but could also improve sealing capacity over time, thereby supporting the long-term integrity of the storage site.

The long-term diffusion of CO₂ and impurities into the caprock was determined to be restricted, with migration distances limited to only a few meters over a span of 10,000 years. This observation corroborates earlier modeling studies that highlight the significance of mineral trapping and self-sealing in mitigating leakage risks. The presence of Fe-, Mg-, and Ca-bearing aluminosilicates within the caprock is crucial to this mechanism, as these minerals facilitate the precipitation of stable secondary phases that lower porosity and permeability.

Thermal effects, although localized, were demonstrated to affect fluid properties and reaction kinetics in the vicinity of the injection well. This underscores the necessity of incorporating thermal coupling in reservoir simulations, particularly when cold CO₂ is injected into formations characterized by low thermal conductivity. Comparable thermal effects have been documented in other research.

Injecting pure CO₂ into a depleted gas reservoir with <30% water saturation causes minimal mineral dissolution, with porosity changes up to 0.1% near the well. CO₂ fills pore spaces and lowers the temperature from 106°C to 30°C, reducing water vapor in the gas phase. In contrast, impure CO₂ (with H₂S, SO₂, NO₂) accelerates dissolution and triggers mineral precipitation, mainly anhydrite/gypsum, leading to porosity reductions of up to 7%, significantly affecting reservoir properties. However, due to the length of the wells, this may not have a detrimental effect on overall injection rates.

This research primarily examines physical and chemical mechanisms; however, it is important to note that microbial processes could also impact the long-term integrity of CO₂ storage. [Bonto et al. \(2022\)](#) emphasize that microbial activities, including biofilm development and mineral changes, may influence both injectivity and sealing characteristics. Although these factors are not modeled in this study, they deserve additional exploration in chalk reservoirs.

While the simulations indicate strong containment and a minimal risk of fault reactivation under the specified conditions, it is essential to highlight that these results depend on keeping injection parameters within the established operational limits. Surpassing these thresholds - whether due to unexpectedly high injection rates, increased impurity levels, or thermal gradients - may result in negative consequences. Subsequent research should investigate these off-envelope situations to enhance risk mitigation strategies.

Conclusion

This research provides a detailed evaluation of the CO₂ storage capacity specific to the Harald East Field, which is a depleted chalk reservoir located in the Danish Central Graben. This study assesses the mechanical, hydraulic, thermal, and chemical reactions of both the reservoir and caprock system under realistic injection scenarios. The modeling framework incorporates analyses at both field and fault scales, calibrated with comprehensive historical data, including production records, 4D seismic data, and subsidence measurements.

The findings indicate that the reservoir can reach its full CO₂ storage capacity, which is limited by the pressure before depletion, while maintaining the structural and stratigraphic integrity of the plume throughout both the injection and post-injection periods. The mechanical response of the reservoir, which includes stress redistribution and compaction, is analyzed through coupled geomechanical modeling, revealing a low probability of fault reactivation under realistic frictional conditions. Simulations of reactive transport suggest that mineral dissolution and precipitation are spatially constrained within the reservoir, and that the longterm diffusion of CO₂ and impurities into the caprock is minimal. The existence of reactive minerals within the seal contributes to a reduction in porosity and may facilitate self-sealing behavior.

These results affirm the technical viability of CO₂ storage in chalk formations, contingent upon a thorough evaluation of site-specific factors such as deformation history, caprock composition, and fault architecture. The modeling methodology established in this study provides a transferable framework for the assessment of other carbonate reservoirs exhibiting similar geological features and enhances the overall understanding of CO₂ dynamics in low-permeability, high-porosity systems.

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